

**University of Colorado at Colorado Springs Department of
Mechanical and Aerospace Engineering**

SPCE 5065: Space Environment Interactions

Summer 2026

Online

Instructor:

Dr. Lynnanne George

Department of Mechanical and Aerospace Engineering

Office: Anschutz 208

lgeorge2@uccs.edu Phone: (719) 235-8610

<https://eas.uccs.edu/staff/lynnane-george>

<https://www.linkedin.com/in/lynnanegeorge/>

Class meetings: TBD

Office hours (online): Mon & Wed 2 - 4

Or by appointment

Course Description:

Introduction to the properties and effects of the environment in which spacecraft and astronauts must operate. Intensive coverage is given to the earth-sun-lunar system. Topics include earth's environment ionosphere, atmospheric chemistry, radiation belts, magnetosphere, aurora, geomagnetic storms, celestial background, and bio-astronautic effects.

Course Introduction:

Space is a place and, like going anywhere, we must prepare. The space environment is harsh and hostile to both spacecraft and humans. To design a successful mission, we must understand how the space environment affects spacecraft performance and lifetime. These interactions can lead to malfunctions, damage, and component or subsystem losses. It is critical to understand the space environment so we can design successful missions.

Of course, not only is the space environment harsh on satellites, but it is also harmful to humans when not carefully planned for. In the final section of this course, we will examine the broad range of environmental stresses on the human element. After completing this course, it will enhance your understanding of the overall space mission design process, and you will be better prepared for your career.

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Expected Course Outcomes:

- Demonstrate base knowledge of the effect of different aspects of the space environment including space weather hazards and associated risks
- Describe solar and space weather phenomena, including aurora, coronal holes, coronal mass ejections (CME), solar flares, sunspots, solar cycle, geomagnetic storms, characteristics of the magnetosphere, and behavior of the interaction between different elements
- Quantify the influence of the space environment on spacecraft and their components
- Design spacecraft robust to space environmental hazards
- Examine bioastronautics risks to humans due to the space environment and mitigation techniques
- Perform detailed analysis to determine effects of space phenomena
- Develop simulations to model space environmental hazards and effects on spacecraft

Course Outline:

The course is divided into roughly six parts:

- The vacuum environment
- Neutral atmospheric particles
- Charged particle and plasma interactions with the spacecraft
- The radiation environment
- Orbital debris
- Human effects

Prerequisites: Graduate students only; SPCE 5005, SPCE 5025

Course Readings:

(CR) Course Readings: All required readings will be provided via .pdf on Canvas. No textbook is required to be purchased. The books below are good references.

(Optional) A.C. Tribble, The Space Environment: Implications for Spacecraft Design, Princeton University Press, 1995.

(Optional) V.L. Pisacane, The Space Environment and Its Effects on Space Systems, AIAA Press, 2016.

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Assessment: The following are the assessment methods and weights for the various items in this course:

Assessment Tool	Percentage of Final Grade
Homework	25%
Test 1	15%
Test 2	15%
Project Milestones	15%
Current Event Presentaiton	5%
Final Project	25%

Grading in the course will follow the schedule below. Note there is no curve.

$93 \leq A \leq 100$	$77 \leq C+ < 79$
$90 \leq A- < 92$	$73 \leq C < 76$
$87 \leq B+ < 89$	$70 \leq C- < 72$
$83 \leq B < 86$	$60 \leq D < 69$
$80 \leq B- < 82$	$F < 60$

Office Hours: Please feel free to stop by during office hours. Also, please make good use of email. I am more than happy to help anytime via email or text.

Canvas will be used for posting homework, solutions, discussions, handouts and any additional materials.

Homework:

Homework will be assigned on a regular basis (approximately 1 per week) and will be posted on Canvas. You may discuss **qualitatively** with other students the concepts and methods required to solve homework problems and to check answers. However, copying, dictating or in any way using the work of another person is **strictly prohibited**. You are required to **independently** prepare the detailed solutions to homework problems. **Each student must turn in an individual/unique problem set** (see policy on Academic Integrity & Cheating). All references for homework must be provided in APA format. Homework is due by 11:59 PM on the on the due date. Late homework will be penalized 10% per day.

Current Event Presentation:

Each student is required to give a 5 – 10-minute presentation on a current event related to the week's material. A sign-up sheet and more information will be provided on the first day of class.

Exams:

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5/22/25

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There will be two tests. Make-up tests will only be scheduled for **extraordinary circumstances** and **with prior written approval** of the instructor, except under clearly documented unavoidable or emergency circumstances.

If you know you will miss any assessment, for any reason, please inform me as early as possible so that other arrangements can be made.

Projects

The project will be individual. There will be milestones to help ensure you are on track.

Expectations

This course requires a minimum of two hours outside of class for every hour in class. This preparation is needed to carefully read the material, practice problems, and complete all required homework and assignments. Attendance at all lectures is expected.

Acceptable and Unacceptable Uses of AI

The use of generative AI tools is permitted in this course for the following activities:

- Brainstorming and refining your ideas
- Fine tuning your research questions
- Finding information on your topic
- Drafting an outline to organize your thoughts
- Checking grammar and style

The use of generative AI tools is not permitted in this course for the following activities:

- Completing group work that your group has assigned to you.
- Writing a draft of a writing assignment
- Writing entire sentences, paragraphs, or papers to complete class assignments

If you use AI, you must include it in your references. An example of how to do that is given here for paraphrasing. Be sure to start the entry with the prompt you used.

Describe the symbolism of the green light in the book *The Great Gatsby* by F. Scott Fitzgerald” prompt. *ChatGPT*, 13 Feb. version, OpenAI, 8 Mar. 2025, chat.openai.com/chat.

References

Reference should be in AIAA format: <https://www.aiaa.org/publications/journals/reference-style-and-format/>

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Dr. George's Introduction

My name is Dr. Lynne George, and I have been teaching in one capacity or another for 27 years. I began my teaching career as a graduate student at Georgia Tech, where I taught an undergraduate Mechanical Measurements course. As a first-generation college graduate, I initially felt uncertain in my new role, an experience that helped me develop empathy for students who may be navigating unfamiliar academic environments.

Following graduate school, I served for 20 years in the United States Air Force, working on space systems, launch vehicles, and satellite programs. I was later selected to teach in the Department of Astronautics at the United States Air Force Academy, where I discovered my passion for teaching and mentoring students. After retiring from military service, I pursued a second career in higher education and now serve as an Associate Teaching Professor in Mechanical and Aerospace Engineering. Throughout my career, I have taught students from a wide range of backgrounds and experiences. These interactions have reinforced my belief that all students can succeed when provided with meaningful opportunities to engage, practice, and grow.

Personal Teaching Philosophy

My teaching philosophy centers on active learning. I believe students learn best when they are engaged in the learning process through problem-solving, design activities, discussion, and other hands-on experiences rather than simply listening to lectures. I also believe that learning requires practice, feedback, and reflection. Students should have multiple opportunities to develop skills before they are formally assessed. By providing guided practice and timely feedback, I strive to build both competence and confidence.

Equally important, students learn best when they feel respected, supported, and valued. Because I was a first-generation college student myself, I am particularly aware that students enter the classroom with different levels of preparation and familiarity with higher education. My goal is to create a learning environment where all students feel comfortable participating, asking questions, and contributing their perspectives.

Implementation of My Teaching Philosophy and Inclusive Practices

I incorporate active learning throughout my courses using collaborative problem-solving, design projects, simulations, discussions, and other activities that encourage students to apply concepts and learn from one another. To foster an inclusive environment, I establish clear expectations for respectful communication and collaboration, encourage participation from all students, and create a classroom culture where questions are welcomed, and mistakes are viewed as part of the learning process. I recognize that students bring diverse backgrounds, experiences, and

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learning styles, and I use a variety of instructional approaches to help all students engage with the material.

I also strive to make expectations transparent and provide multiple opportunities for students to practice and demonstrate their learning. Through accessible office hours, mentoring, and regular feedback, I work to ensure that students feel supported while maintaining high academic standards.

Finally, I continually reflect on my teaching and seek feedback from students and colleagues. I regularly participate in learning opportunities and courses to improve my own teaching. By remaining responsive to student needs and committed to improvement, I aim to create a learning environment where every student feels challenged, supported, and capable of success.

Disability Services

If you are a student with a disability and believe you will need accommodations for this class, it is your responsibility to register with Disability Services and provide them with documentation of your disability. They will work with you to determine what accommodations are appropriate for your situation. To avoid any delay, you should contact Disability Services as soon as possible. Please note that accommodations are not retroactive and disability accommodations cannot be provided until a Faculty Accommodation Letter has been given to me. Please contact Disability Services for more information about receiving accommodations at Main Hall, 3rd floor, room 324, 719-255-3354 or dservice@uccs.edu.

Veteran and Military Affairs

If you are a military student with the potential of being called to military service or training during the course of the semester, you are encouraged to contact your UCCS course instructor no later than the first week of class to discuss the class attendance policy. Please see the Veteran and Military Affairs website for more information.

Student Support Network Alerts

I participate in Student Support Network, a UCCS initiative designed to encourage student success. If your success in this course is at risk, you may receive an alert email about your attendance, assignment grades, or overall performance. Please pay attention to these emails and take the recommended actions. They are sent to help you be successful!

Key facts about the Support Network:

- The main goal of this is to help you be successful during your time at UCCS.
- Alerts can be viewed only by the instructor who submitted the alert and student support team.

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What to do if you receive an alert?

- Don't panic! Read the email thoroughly and act. Follow the advice and directions provided in the email.
- Meet with your instructor, academic advisor, or other student support team member as soon as possible.

Mental Health and Wellbeing

Mental health challenges can interfere with academic performance. The source of symptoms might be related to your course work; if so, please speak with me. However, problems with other parts of your life can also contribute to decreased academic performance. If you are unsure if you need assistance, free online mental health screenings can be found here.

If you are experiencing a mental health emergency (i.e., you do not feel physically safe), please immediately contact one of the resources listed below:

- **Call UCCS Public Safety at 719-255-3111, call or text the National Suicide & Crisis Lifeline at 988, call 911, or go to the nearest emergency room.**

Remember, we care about your wellbeing, so if you are struggling (even if this is not an emergency) please reach out for help.

For confidential mental health services, visit the Wellness Center located inside the Gallogly Recreation and Wellness Center. The first 3 visits are free! Phone: 719-255-4444. Hours: Monday–Friday, 8a.m.–5p.m. You can also contact Telus Health Student Support for free, confidential 24/7 mental health and wellbeing support through the Telus Health app or visit our free mental health and resilience digital platform. Resource information can be found at 24/7 Mental Health Resources | Gallogly Recreation and Wellness Center.

If you need a community resource, Diversus Health Lighthouse has a Walk-In Crisis Center at 115 S. Parkside Drive, Colorado Springs, CO 80910, Hotline: 844-493-8255, Office: 719-572-6100. Other community resources can be found at Resources in the Colorado Springs Community | Gallogly Recreation and Wellness Center.

Accommodations based on Religion or Creed

UCCS complies with federal regulations regarding requests for accommodation by students and employees for observation of their religion or creed. Students may contact the Office of the Dean of Students at dos@uccs.edu to discuss academic accommodation requests. Employees, including student employees, may contact the Office of Human Resources for workplace accommodation requests at hrhelp@uccs.edu.

Professional Behavior in the Classroom

This course employs evidence-based approaches and scientific methodologies to examine the space environment. While students may hold various personal beliefs and worldviews, our

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academic discussion will focus on analyzing evidence, evaluating research methods, and engaging with peer-reviewed scholarship. All students are expected to engage with course materials and participate in discussions in a respectful, analytical manner, even when the content challenges their existing views. Different perspectives are welcome when expressed constructively and supported by reasoned arguments.

Responsible Reporting: Institutional Equity / Title IX

The Office of Institutional Equity (OIE) administers the University of Colorado (CU) Sexual Misconduct, Intimate Partner Violence, and Stalking Policy (APS 5014), the CU Protected Class Nondiscrimination Policy (APS 5065), and the Conflict of Interest in Cases of Amorous Relationships Policy (APS 5015). All UCCS faculty and staff, including student employees, are considered “responsible employees” and must report any suspected or known conduct that may be in violation of these policies (“prohibited conduct”) directly to the OIE (mandatory reporting). Responsible employees must also provide the OIE’s contact information and information about how to file a report of prohibited conduct to anyone who has disclosed alleged prohibited conduct. This includes reports made to the employee by others, regardless of the time or location of the alleged incident. This reporting requirement exists to support the safety of our campus community, and there is no statute of limitations. Prohibited conduct includes unwanted verbal or physical conduct that is sexual in nature, or based on a protected class (such as, sex, race, veteran status, disability, etc.). You may review the Applicable Policies and OIE’s Resolution Procedures. While reporting prohibited conduct to the OIE is required by UCCS employees, participation in an OIE resolution process is voluntary. Confidential reporting, and access to medical and mental health services, are available to students through the UCCS Wellness Center at 719-255-4444. See video training on relevant topics.

Title IX Coordinator: Laura Emmot lemmot@uccs.edu 719-255-3725.

UCCS Procedures for Assisting Students and Employees who are Pregnant, Nursing, or with Pregnancy Related Conditions. All employees to whom a student discloses pregnancy or pregnancy related conditions are required to provide the student with OIE’s contact information and share that the Title IX Coordinator can assure accommodations and resources. OIE can be contacted at (719) 255-4324 or equity@uccs.edu. Procedures are posted here.